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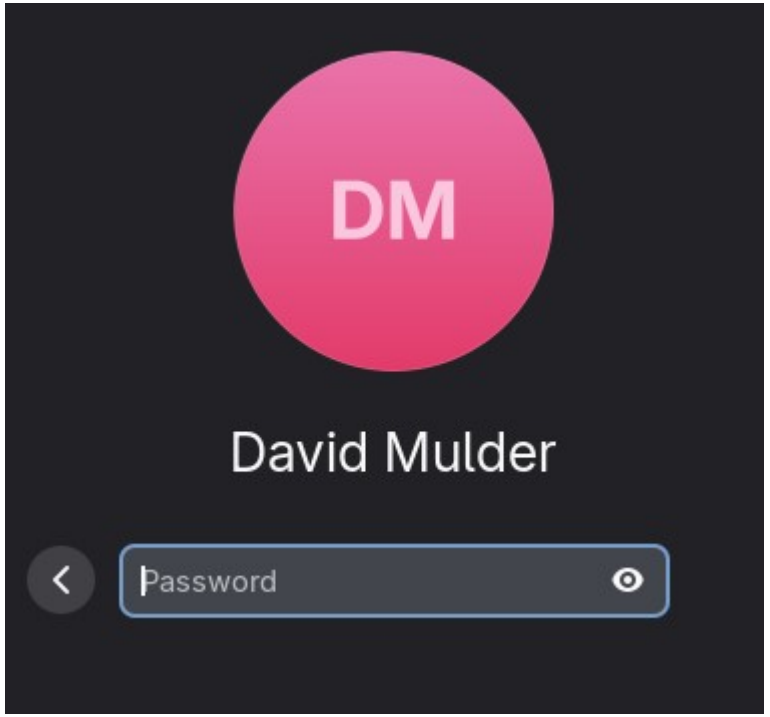
13 DECEMBER 2025

Who Authenticates Linux?

Rethinking PAM & NSS in the Age of Cloud Identity

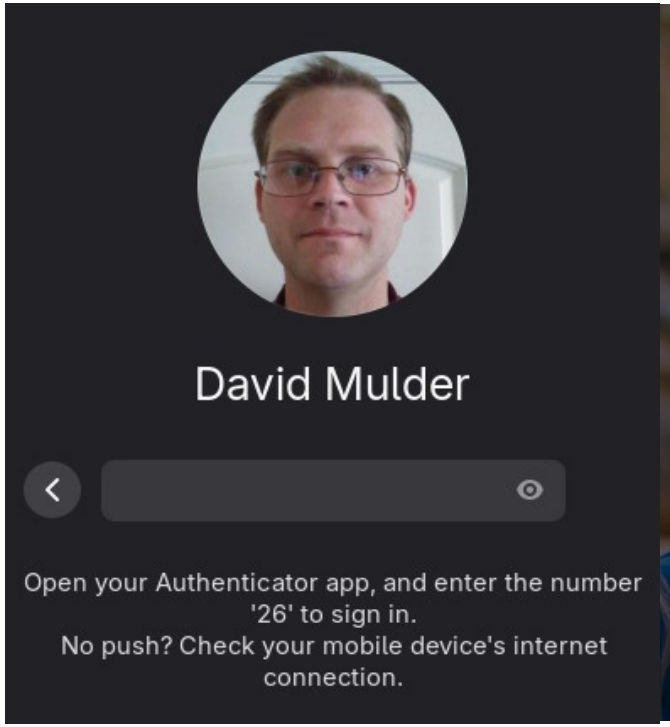
Authentication, Then vs Now (real examples)

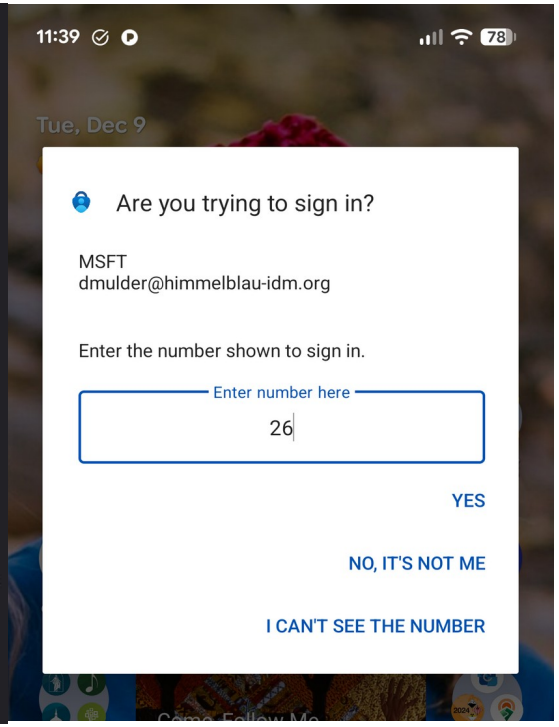
PASSWORD (in-band) via pam_unix



Type a secret

MFA (out-of-band) via Himmelblau





Approve elsewhere

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Where Modern Identity Breaks Linux

- PAM exposes only simple interaction primitives
- Authentication flows are strictly linear (no branching, no backtracking)
- Modern IdPs **assume** a browser is **always** available

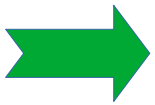
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What a Standardized MFA API Could Look Like

From workarounds → Protocols

Today (bespoke, fragile)

- Reverse-engineered IdP flows
- Custom polling & token exchange
- Device Authorization Grants (RFC8628)
  - Is this Enterprise-grade security, or a hacked-up SmartTV flow?

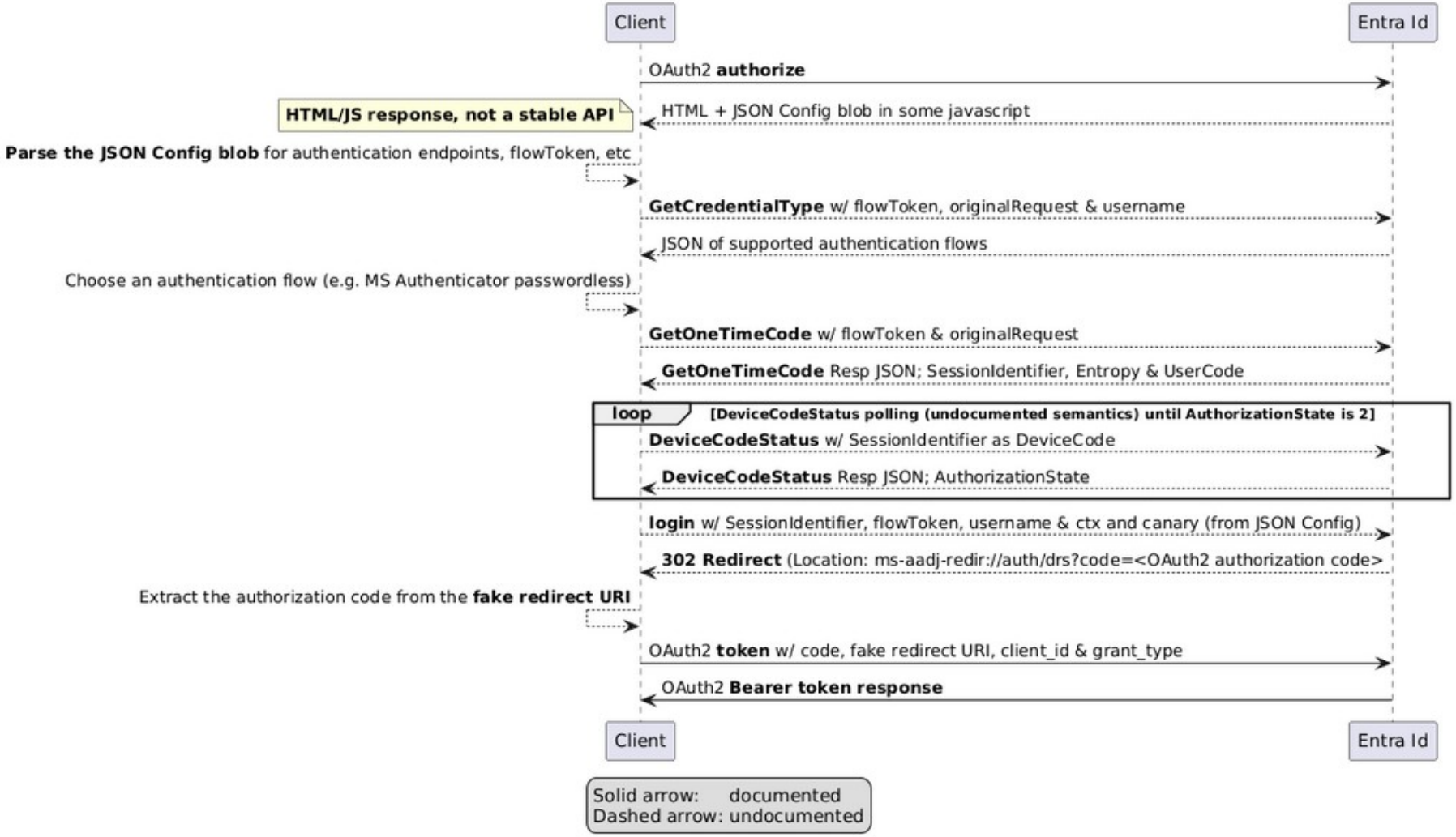


Desired (standardized)

- **Browser-independent MFA APIs**
- Explicit flow types (polling, FIDO/WebAuthn, push, auth code)
- Structured choices & transitions
- Works in SSH, GDM, services

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Here’s how we’re surviving today



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Why Device Authorization Grant Isn't Enough — and What's Missing

Device Authorization Grant (RFC 8628)

- Designed for "smart TVs, media consoles, digital picture frames, and printers" (RFC 8628 Abstract)
- Works as a fallback, not as a foundation

What's Missing

- A browser-independent MFA API
- Structured method discovery & selection
- Explicit, machine-readable state transitions

Open Questions

- Where should this be standardized?
- OAuth2 extension? New IETF work? Something else entirely?

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Thank You

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